

Worksheet 7D: Markov Chains — Probability Meets Matrices — Solutions

Series 7: Linear Algebra
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

Instructor Copy — Not for Distribution

Note to instructor: Key answers are given below. Accept any mathematically equivalent form unless the question specifies otherwise.

Part A

1. (a) 0.6 (b) 0.4 (c) Row 1: $0.7 + 0.3 = 1$; Row 2: $0.4 + 0.6 = 1$. ✓
2. (a) 0.5 (b) 0.2 (c) All rows sum to 1. ✓
3. $P = \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$ (row 1: from studying; row 2: from procrastinating)
4. Valid: (a), (c), (d). Invalid: (b) sum = 1.1; (e) negative entry.
5. $\pi^{(1)} = [0.4, 0.6]$; $\pi^{(2)} = [0.52, 0.48]$
6. $\pi^{(1)} = [0.55, 0.45]$; $\pi^{(2)} = [0.565, 0.435]$
7. Two days after a definite rainy day, there is about a 52% chance of sunshine and 48% chance of rain.
8. With the column convention we write $\pi^{(t+1)} = P^T \pi^{(t)}$ with column state vectors; answers are the same, just transposed.
9. $\pi^{(1)} = [0.8, 0.2]$; $\pi^{(2)} = [0.76, 0.24]$
10. (a) $P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ (b) Past flips give no information about the next flip — the Markov property is trivially satisfied.

Part B

11. $P^2 = \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix}$
12. 0.61
13. $[1 \ 0] \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix} = [0.61, 0.39]$. ✓
14. $\pi^{(1)} = [0.7, 0.3]$; $\pi^{(2)} = [0.7(0.7) + 0.3(0.4), 0.7(0.3) + 0.3(0.6)] = [0.61, 0.39]$. ✓
15. Row $n = 2$: $[0.610, 0.390]$; $n = 3$: $[0.583, 0.417]$; $n = 4$: $[0.575, 0.425]$

16. The distribution converges toward approximately $[0.571, 0.429]$.
17. Starting rainy: $n = 2$: $[0.52, 0.48]$; $n = 3$: $[0.556, 0.444]$; $n = 4$: $[0.567, 0.433]$. Both starting points converge to the same values.
18. Long-run predictions are independent of the starting state — the system “forgets” its initial condition.

Part C

19. $\pi = [4/7, 3/7] \approx [0.571, 0.429]$
20. $[4/7, 3/7] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = [(4/7)(0.7) + (3/7)(0.4), (4/7)(0.3) + (3/7)(0.6)] = [4/7, 3/7]$.
✓
21. Yes — the table values in problem 15 approach $[0.571, 0.429]$.
22. $[\pi_S, \pi_P] \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix} = [\pi_S, \pi_P]$ and $\pi_S + \pi_P = 1$. This gives $0.8\pi_S + 0.6\pi_P = \pi_S$, i.e. $0.6\pi_P = 0.2\pi_S$, so $\pi_S = 3\pi_P$. With $\pi_S + \pi_P = 1$: $\pi_S = 3/4$, $\pi_P = 1/4$.
23. In the long run, the student spends 75% of the time studying and 25% procrastinating.
24. The stationary distribution reveals what fraction of time (or visits) is spent on each page in the long run — directly useful for understanding user behaviour.
25. (a) Yes, all paths eventually enter state 1 and stay. (b) $\pi = [1, 0]$ (completely absorbed into state 1).
26. $P = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}$. In the long run, the walker is absorbed into either L or R; the exact probability depends on starting position.
27. A page linked by other high-ranked pages gets many “votes.” In the stationary distribution, each page’s score equals a weighted sum of the scores of pages linking to it — exactly what the eigenvector captures.
28. All rows are $[\pi_1, \pi_2, \dots]$ already; the chain reaches stationarity immediately. Degenerate: any initial state gives the same distribution on the next step.

Part D

29. (a) $P = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 1 & 0 & 0 \\ 0.5 & 0.5 & 0 \end{bmatrix}$ (b) Each row sums to 1. ✓ (c) Page A likely ranks highest: it receives links from both B and C.
30. (a) “like” (probability 0.5) (b) $P(I) \cdot P(\text{like}|I) \cdot P(\text{cats}|\text{like}) = 1 \times 0.5 \times 0.6 = 0.3$ (c) Estimated from counts in a large text corpus.
31. Stationary distribution is $[0.25, 0.25, 0.25, 0.25]$ — uniform. Long-run DNA composition is 25% of each base.

32. (a) $1 - 0.92 = 0.08$ or 8% (b) 0.30 or 30% (c) An inactive person has a 15% probability of becoming unemployed (rather than employed or remaining inactive) in the next year.
33. Chains with large diagonal entries (“sticky” states) mix slowly — the chain stays in each state for a long time before transitioning.
34. The Markov property ensures the value function and policy depend only on the current state, making dynamic programming tractable.
35. Ergodic: from state 1 to state 3 via $1 \rightarrow 2 \rightarrow 3$; from 3 to 1 via $3 \rightarrow 1$ directly. Yes, ergodic.
36. (a) $\boldsymbol{\pi}^{(t+1)} = \boldsymbol{\pi}^{(t)}P$ is exactly a row vector times a matrix. (b) Basic probability gives $P(A)$; Markov chains give $P(\text{state at time } n)$ for sequences. (c) $\boldsymbol{\pi}P = \boldsymbol{\pi}$ can be written $\boldsymbol{\pi}(P - I) = \mathbf{0}$, i.e. $\boldsymbol{\pi}$ is a left eigenvector of P with eigenvalue 1.